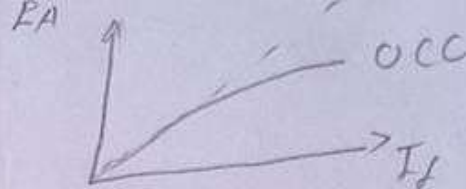


$$Q = 3 V_L I_a \cos \theta$$

$$V_L = \sqrt{3} V_T$$



**CAT II & III: 3rd YEAR SECOND SEMESTER, BSc
MECHATRONIC ENGINEERING
EMT 3204: ELECTRICAL MACHINE DRIVES II**

Q1. Explain the difference between a Single excited machine and Doubly excited machine [2 marks]

Q2. With an aid of a diagram, explain the power flow in an induction motor. [5 marks]

Q3. A 480-V, 60-Hz, 50-hp, three-phase induction motor is drawing 60 A at 0.85 PF lagging. The stator copper losses are 2 kW, and the rotor copper losses are 700 W. The friction and windage losses are 600 W, the core losses are 1800 W, and the stray losses are negligible. Find the following quantities:

- | | |
|-------------------------------------|-----------|
| I. The air-gap power, P_{AG} | [2 marks] |
| II. The power converted, P_{conv} | [2 marks] |
| III. The output power, P_{out} | [2 marks] |
| IV. The efficiency of the motor | [2 marks] |

Q4. Discuss three basic approaches that can be used to safely start a synchronous motor. [9 marks]

Q5. State and explain the two different types of induction motor rotors [3 marks]

Q6. Using phasor diagrams, explain the behavior of synchronous motors under varying conditions of Field current. [6 marks]

$$Q_{out} = V_L I_L \sqrt{3} \sin \theta$$

$$= 3 V_\phi I_a \sin \theta$$

P-1AG-7

$$N_s = \frac{120f}{P}$$

$$P = \sqrt{3} V_T I_L \cos \theta$$

$$\omega_m = \frac{2\pi N_s}{60}$$

$$1 \text{ hp} = 745.7 \text{ W}$$

$$P_{in} = T \cdot \omega_m$$

$$N_s = \frac{120f}{P}$$

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$$V_\phi = E_A - I_a R_a - jX_s I_a$$

Q6. A 208-V, 45-hp, 0.8-PF-leading, Delta-connected, 60-Hz synchronous machine has a synchronous reactance of 2.5 ohms and a negligible armature resistance. Its friction and windage losses are 1.5 kW, and its core losses are 1.0 kW. Initially, the shaft is supplying a 15-hp load, and the motor's power factor is 0.80 leading.

- Sketch the phasor diagram of this motor, and find the values of I_A , I_L and E_A . [4 marks]
- Assume that the shaft load is now increased to 30 hp. Sketch the behavior of the Phasor diagram in response to this change. [3 marks]
- Find I_A , I_L and E_A after the load change. What is the new motor power factor? [4 marks]

Q7. The following test data were taken on a 7.5-hp, four-pole, 208-V, 60-Hz, design A, Y-connected induction motor having a rated current of 28 A. Sketch the per-phase equivalent circuit for this motor. [16 marks]

DC test:	$V_{dc} = 13.6 \text{ V}$	$I_{dc} = 28.0 \text{ A}$
No-load test:	$V_r = 208 \text{ V}$ $I_a = 8.12 \text{ A}$	$f = 60 \text{ Hz}$ $P_m = 420 \text{ W}$
Locked-rotor test:	$I_a = 8.20 \text{ A}$ $I_c = 8.18 \text{ A}$	$V_r = 25 \text{ V}$ $I_a = 28.1 \text{ A}$ $I_b = 28.0 \text{ A}$ $I_c = 27.6 \text{ A}$
		$f = 15 \text{ Hz}$ $P_m = 920 \text{ W}$

$$Z_{14.7} = X_1 + X_m$$

$$P_{out} = \sqrt{3} V_o I_L \cos \theta$$

$$P_{conv} = 3 E_A I_a \cos \theta$$

$$P_{out} = \frac{3 E_A V_o \sin \delta}{X_s}$$